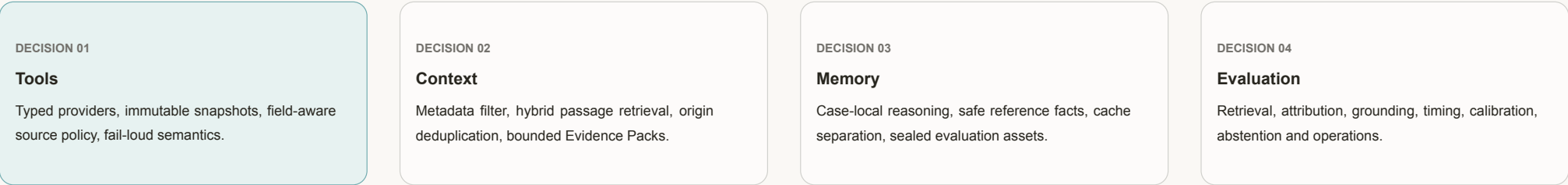
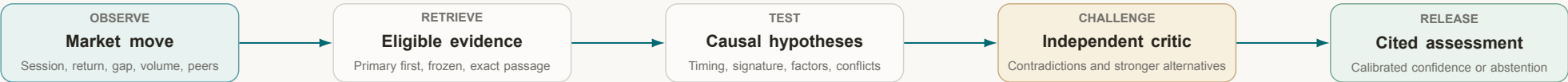


ASX Unusual Trading Investigation Agent

A point-in-time, evidence-grounded system for explaining unusual moves in ASX-listed equities.

Deterministic market truth constrains model reasoning. Evidence and calibration determine what may be claimed.

Input: an ASX ticker and an ASX date. Output: a confidence-rated explanation in AUD and AEST/AEDT, with every material claim linked to evidence and every limitation visible.

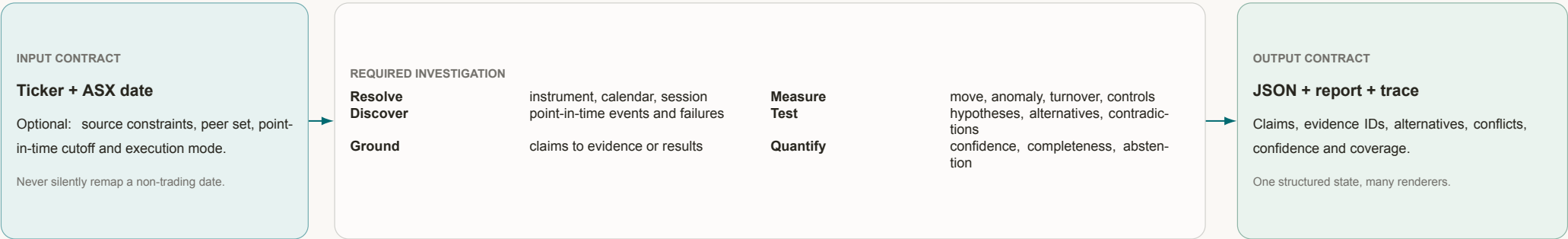


STATUS BOUNDARY This dossier specifies the target architecture. The repository currently contains a committed design baseline and an untracked scaffold; it is not a completed agent and has no evaluation results.

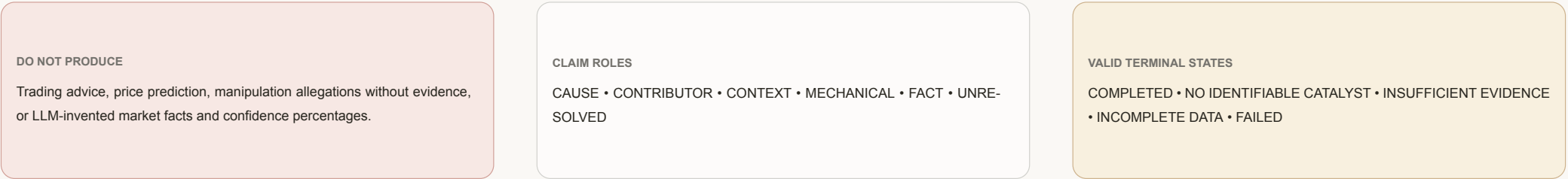
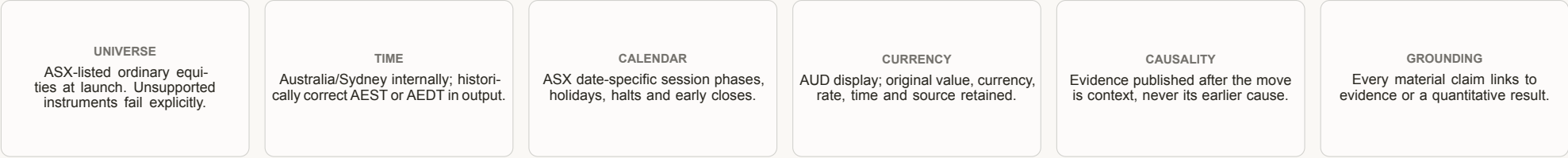
Basis: repository design specification, master development plan, frontend design baseline and verified repository inspection [R9-R12].

What the product must prove

The answer must be inspectable, reproducible and allowed to end in uncertainty.



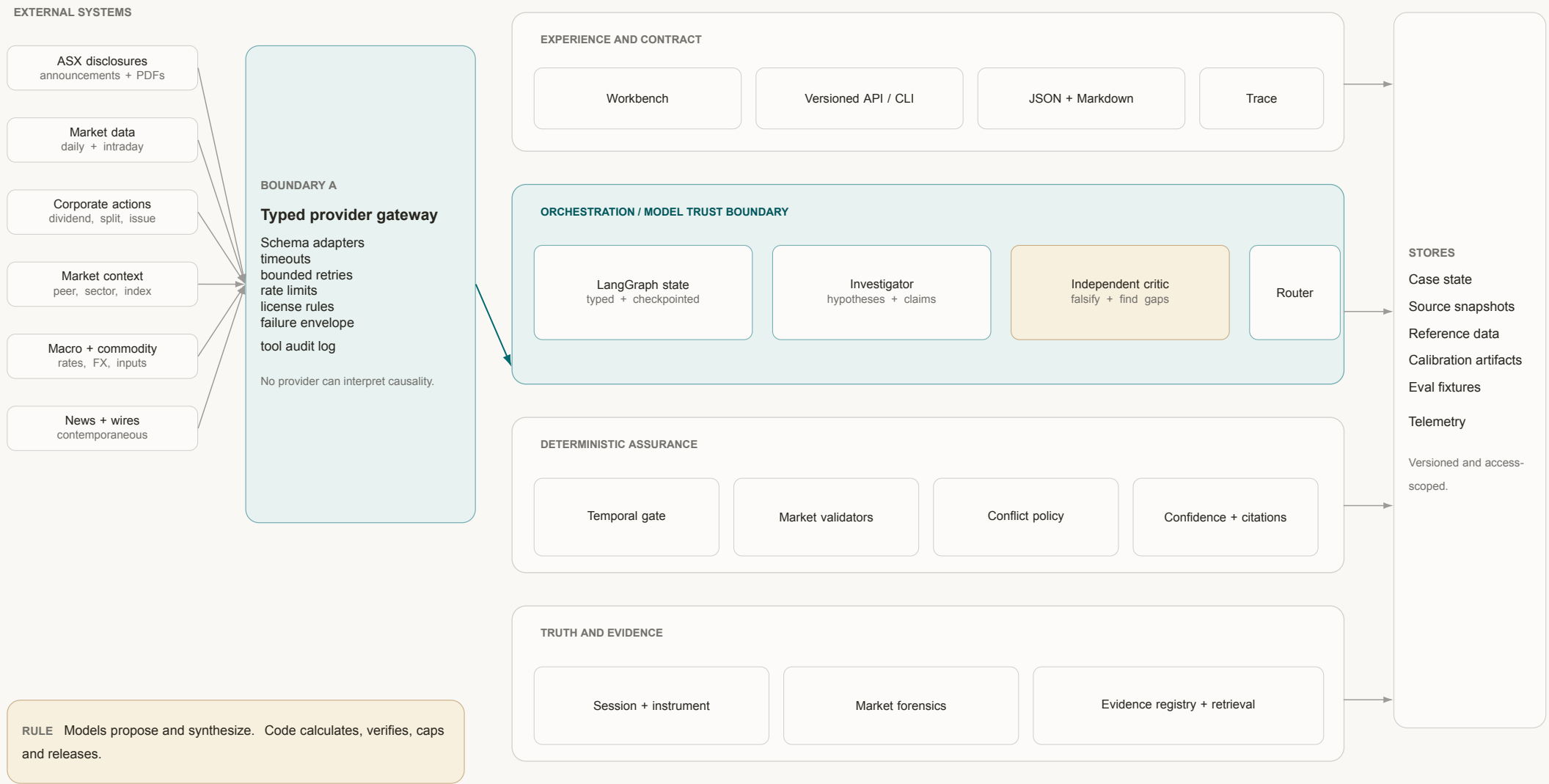
Non-negotiable constraints



ASX session phases use Sydney time; normal continuous trading ends at 16:00 before the closing auction [R1-R2]. Continuous disclosures are primary issuer evidence [R3].

Small agent surface, strong assurance core

Two reasoning nodes operate inside deterministic market, evidence and release boundaries.

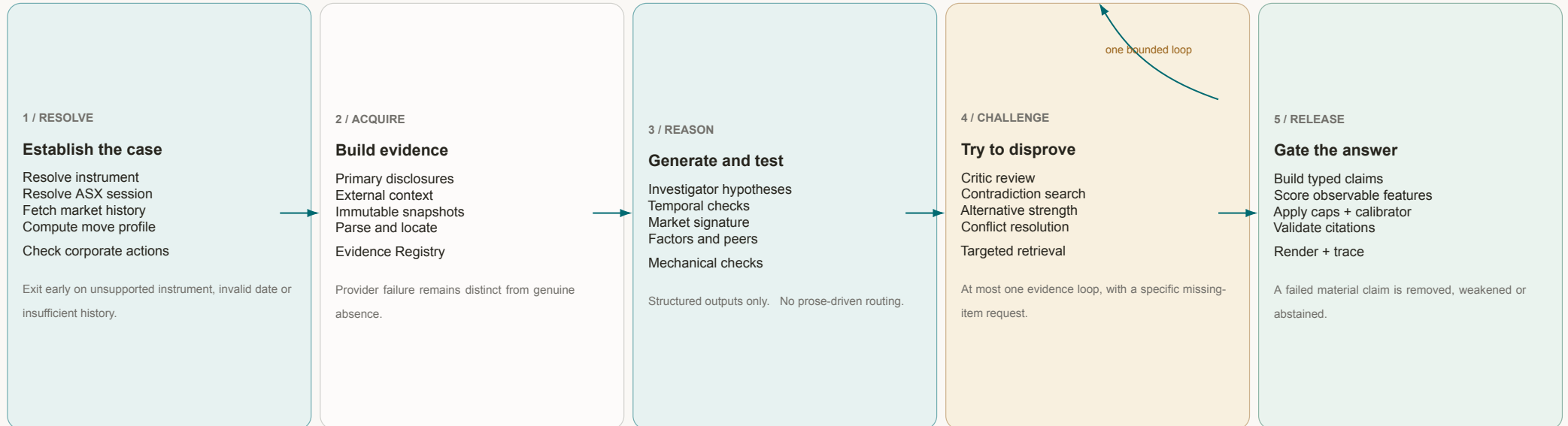


Architecture baseline [R9-R10]. LangGraph checkpoints support versioned graph-state persistence and resumption [R8].

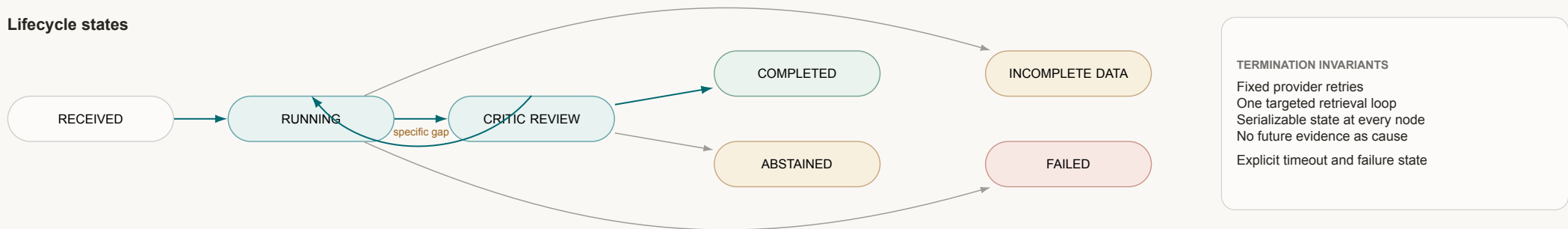
Investigation graph and state machine

A bounded workflow separates discovery, validation, criticism and release.

UML activity flow



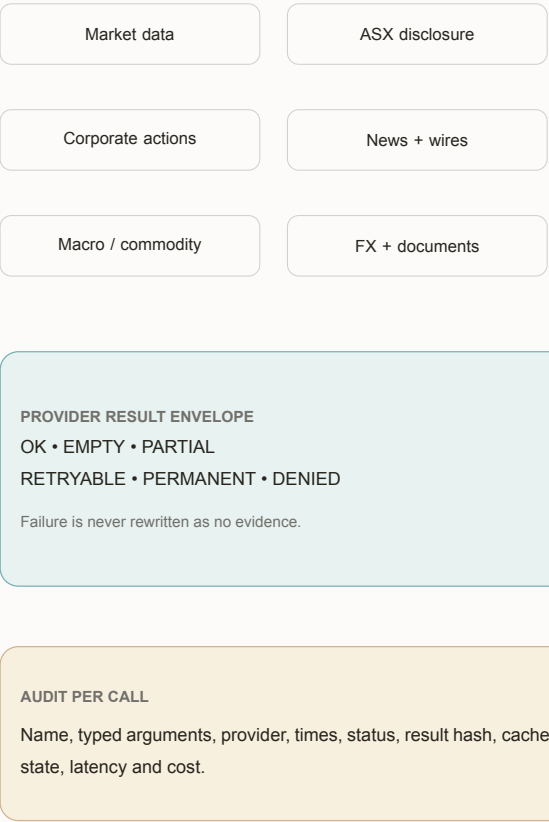
Lifecycle states



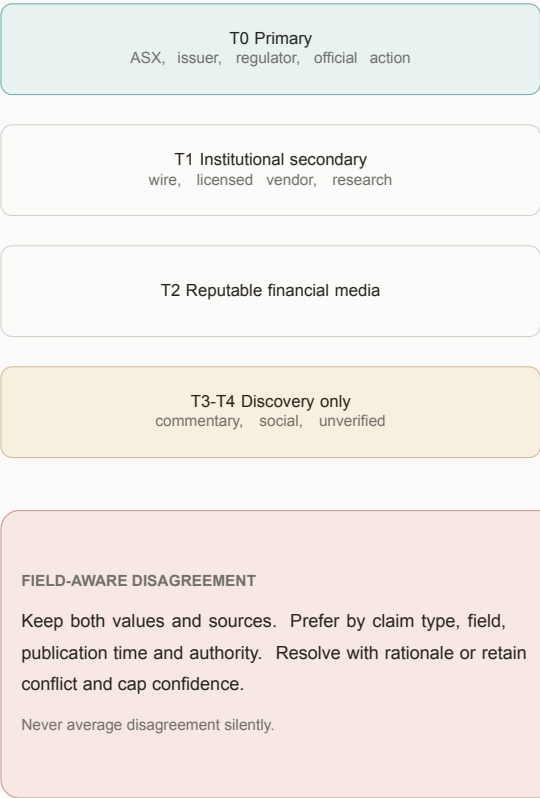
From provider call to bounded Evidence Pack

Data enters through typed contracts; documents remain untrusted until normalized, filtered and registered.

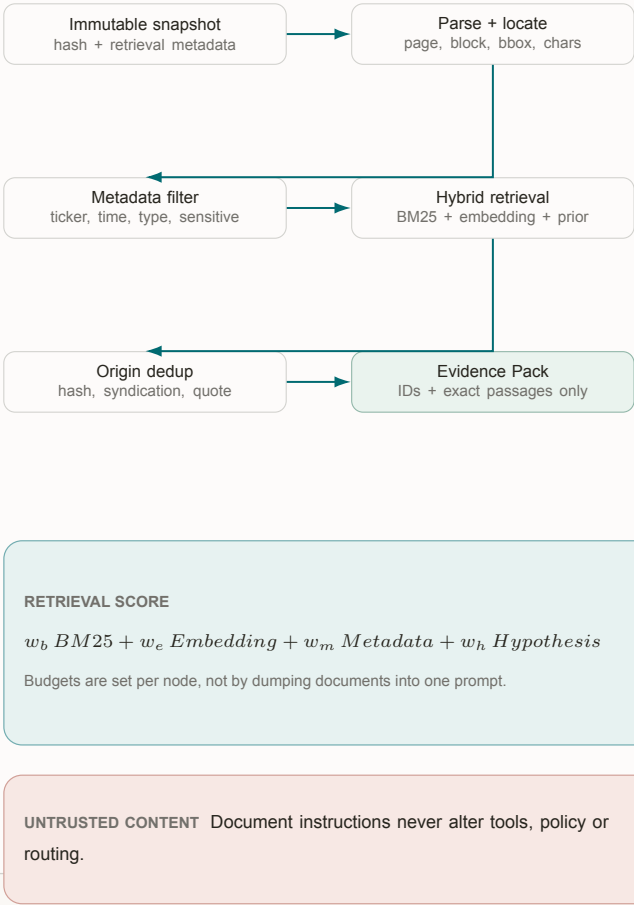
A. Typed tools



B. Source and conflict policy



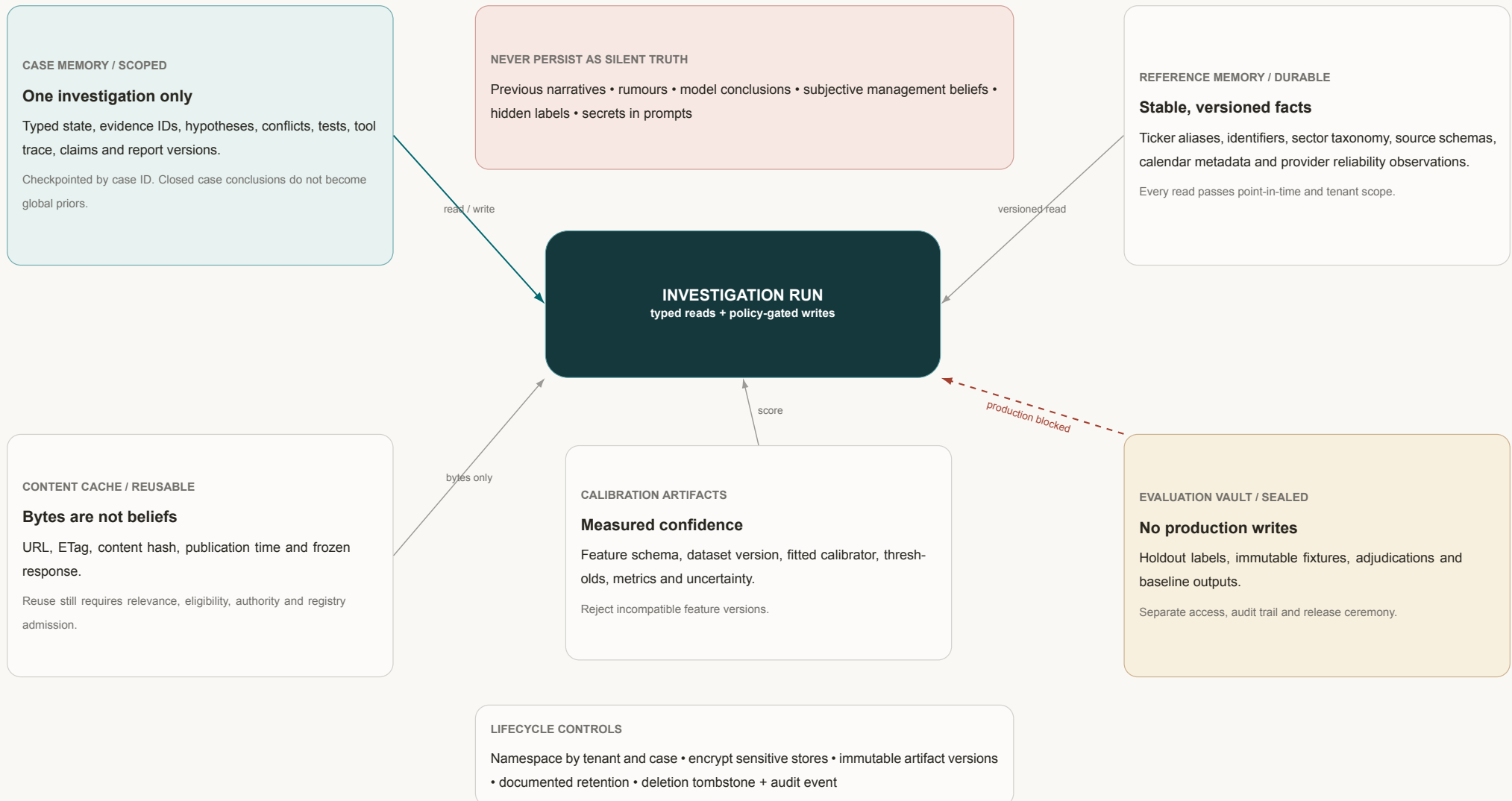
C. Context pipeline



ASX disclosures and session sources [R1-R3]. Indirect prompt injection can arrive through retrieved content, tool output and memory [R6]. Architecture baseline [R9-R10].

Persist facts and artifacts, not conclusions

Memory is split by scope, authority and lifecycle so one case cannot silently bias the next.

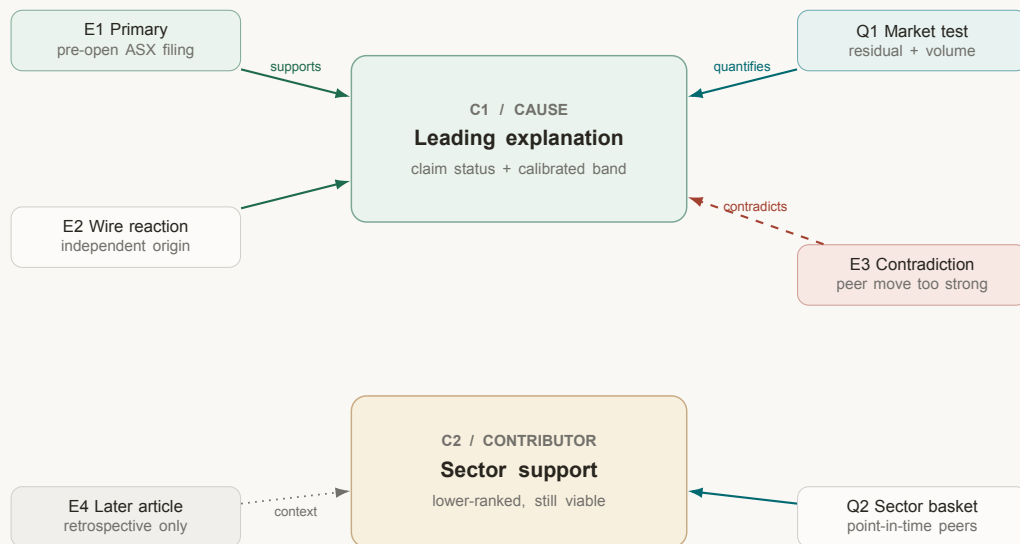


Memory boundaries derive from the project baseline [R9-R10]. LangGraph distinguishes checkpointed thread state from durable stores [R8]. NIST calls for lifecycle risk controls [R4-R5].

Confidence is earned by observable evidence

The claim-evidence graph is the source of truth; the report is only its rendering.

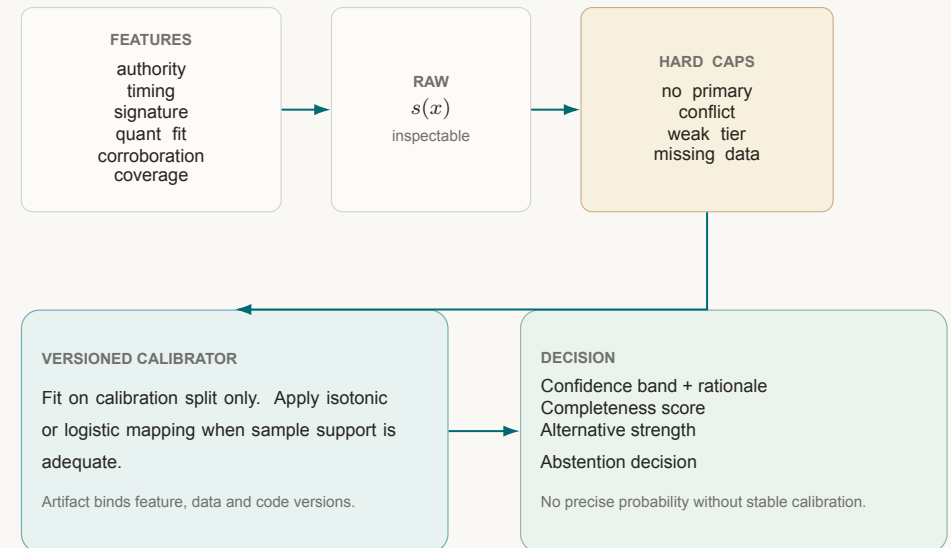
Typed claim-evidence graph



RELEASE RULE

A material claim with no registered support cannot ship. Retrospective evidence cannot support an earlier cause. Syndicated copies count as one origin. Conflicts remain visible.

Confidence and abstention pipeline



KEEP FOUR QUANTITIES SEPARATE

Anomaly score • claim support
Selected-hypothesis confidence • investigation completeness

A dramatic move can have weak attribution. Complete retrieval can still yield no catalyst.

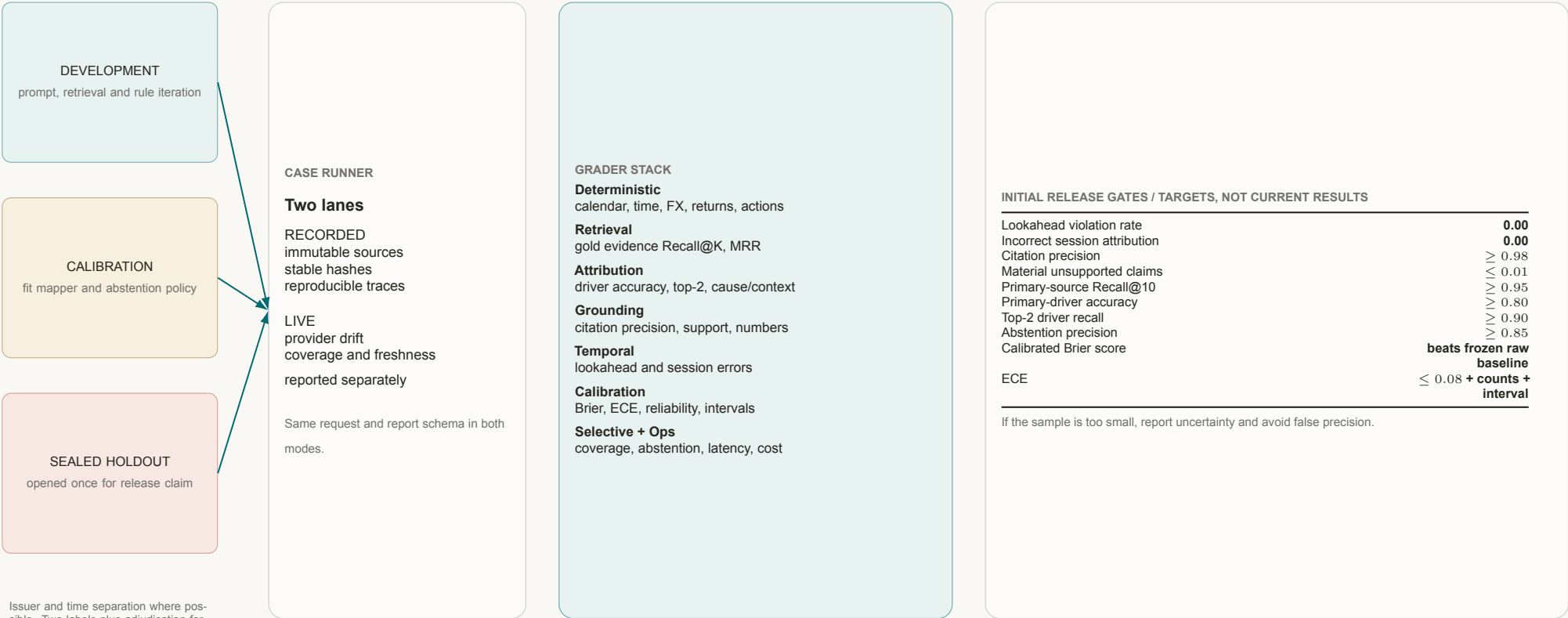
$$BS = \frac{1}{N} \sum_i (p_i - y_i)^2 \quad ECE = \sum_b \frac{|S_b|}{N} |acc(S_b) - conf(S_b)|$$

Calibration test: predictions near 0.80 should be correct about 80% of the time in comparable cases.

Measure every failure surface separately

The harness must show whether a failure came from data, retrieval, reasoning, grounding, time or calibration.

Evaluation architecture



Issuer and time separation where possible. Two labels plus adjudication for causal gold.

FAILURE TAXONOMY

DATA • RETRIEVAL • TEMPORAL • QUANTITATIVE • REASONING • GROUNDING • CALIBRATION • INFRASTRUCTURE

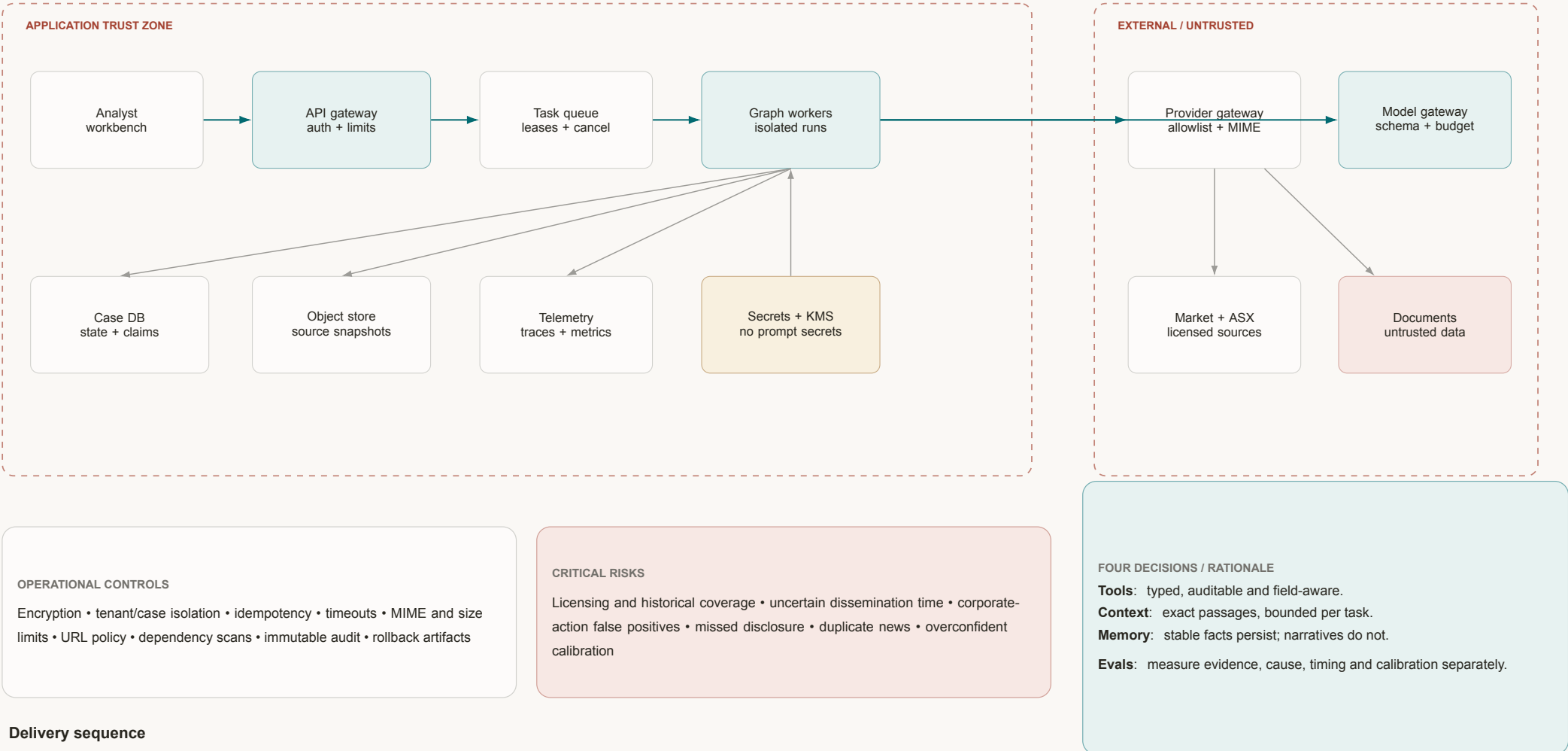
Every failed case links to its trace and the first failing stage.

Current repository: no eval dataset, harness or measured result is available.

Deployment, controls and delivery path

The production boundary protects source integrity, model access, case data and the release decision.

Deployment and trust boundaries



Sources: governance [R4-R5], security [R6], checkpoints [R8], roadmap [R9-R12].